

Polaris — Fleet Certification Evidence Package

Catalog v1.0.0 · Generated 2026-07-21T01:33:57Z · Fleet status: **GREEN**

Subsystem	Status
BATTERY_PACK	GREEN
FIXED_WING_UAV	GREEN
LANDING_MULTIROTOR	GREEN

Requirements traceability

ID	Requirement	Bound	Value	N	Status	Sev	Hash
FC-BAT-001	EKF SOC RMS error under worst characterisation scenario	<= 0.06	0.05461	6	PASS	MAJOR	b3269fdce799
FC-BAT-002	EKF advantage over coulomb counting	>= 60.0	90.06	6	PASS	MAJOR	b3269fdce799
FC-BAT-003	Failure-mode triage identifies at least one degradation cluster	>= 3.0	6	8	PASS	MAJOR	260596e970ad
FC-BAT-004	Remaining useful life is finite for every non-healthy mode	==True 1.0	yes	4	PASS	CRITICAL	f8abd84ce8a2
FC-UAV-001	Nominal mission completes with full test-card compliance	pass_frac>= 110		4	PASS	CRITICAL	3981f6f1b2f6
FC-UAV-002	Geofence containment across every scenario	==False 0.0	no	32	PASS	CRITICAL	3981f6f1b2f6
FC-UAV-003	Cross-track RMS bound in nominal conditions	<= 5.0	1.261	4	PASS	MAJOR	3981f6f1b2f6
FC-UAV-004	Anomaly triage per-class accuracy	>= 0.85	1	1	PASS	MAJOR	0672ecb0bd53
FC-LDG-001	Touchdown lateral error inside pad acceptance radius (nominal)	<= 0.5	0.03884	12	PASS	CRITICAL	d47c6eefc429
FC-LDG-002	Sink rate at touchdown (nominal)	<= 1.0	0.3825	12	PASS	CRITICAL	d47c6eefc429
FC-LDG-003	Unsafe finals must trigger a go-around, not a landing	==True 1.0	yes	24	PASS	CRITICAL	d47c6eefc429
FC-LDG-004	Vision availability on final	pass_frac>= 019		12	PASS	MAJOR	d47c6eefc429
FC-LDG-005	Triage identifies each fault family it saw	>= 8.0	9	10	PASS	MAJOR	32772c91a7c1

Artefact manifest (SHA-256, first 12 chars)

Subsystem	Role	Path	Hash	Present
BATTERY_PACK	estimator_benchmark	estimator_benchmark.json	b3269fdce799	yes
BATTERY_PACK	triage_model	triage_model.json	260596e970ad	yes
BATTERY_PACK	rul_projections	rul_projections.json	f8abd84ce8a2	yes
BATTERY_PACK	cycle_records	cycle_records.json	241526c7e703	yes
FIXED_WING_UAV	campaign	campaign.json	3981f6f1b2f6	yes
FIXED_WING_UAV	triage	triage.json	0672ecb0bd53	yes
LANDING_MULTIROTOR	campaign	campaign.json	d47c6eefc429	yes
LANDING_MULTIROTOR	triage	triage.json	32772c91a7c1	yes

All source telemetry is synthetic simulator output. This package records what the sprint 7–9 test-card engines emitted for the artefact bytes hashed above; it does not claim provenance for any physical hardware.